

Deepak Mallampati

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Kent, OH (open to NYC relocation) -- F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production **Gen AI workflows and LLM systems** end-to-end in regulated environments. Customer-embedded delivery experience: **Healthcare RAG** (~35% retrieval precision, >90% grounding alignment), **Agentic Healthcare Claims** multi-agent pipeline with schema-validated JSON contracts (audit-ready, hallucination drift eliminated), **predictive ML serving** with FastAPI/Docker (~30% defect reduction). Comfortable iterating directly with non-technical stakeholders, translating operational friction into production AI. Strong Python + TypeScript breadth; ML fundamentals (XGBoost, walk-forward validation, threshold calibration); HIPAA-conscious governance discipline transferable to enterprise verticals.

CORE COMPETENCIES

Production Gen AI Workflows . Customer-Embedded Engineering . LLM Orchestration & Multi-Agent . Production RAG & Grounding . ML Foundations (Eval, Training, Decomp) . Python / Java / TypeScript / C++ . FastAPI / Docker / REST APIs . Regulated-Domain Governance . End-to-End Production Delivery

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee -- USA | Healthcare Technology, AI Consulting

- Built **production RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved precision ~35% and reduced irrelevant retrieval >30%; response alignment exceeded 90%.
- Built **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent stability ~25% and reduced hallucination rate >30%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; class weighting + stratified sampling + threshold calibration improved recall 15-20% on high-risk while holding precision >90%.
- Packaged LLM/ML inference as **FastAPI/Flask** RESTful services containerized with Docker; structured logging + load simulation cut post-deployment defects ~30%.
- Maintained HIPAA-conscious data governance and authored stakeholder-facing system-limitation docs — direct analog to enterprise-customer-embedded AI delivery in regulated industries.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India | Enterprise ERP, Oil & Gas

- Optimized T-SQL stored procedures powering the Synthesis Order-to-Cash ERP platform; reduced query execution ~20%, retrieval latency ~25%.
- Built CI/CD pipelines with Jenkins for schema + stored procedure deployments; reduced deployment errors >30%, improved release cycle ~35-40%.
- Designed Grafana observability dashboards for production SQL systems; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.

Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer -- Hyderabad, India | Nonprofit Tech, Applied NLP

- Built transformer-based video summarization for 5,000+ sessions; reduced manual review time 60-70%; deployed as Flask API for non-technical users.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~30% and raised contextual accuracy >85%.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent Gen AI pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations — direct analog to AIP-style production Gen AI workflows. RAG-grounded CPT/ICD validation against vector-indexed policy docs; ANN similarity for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, vector search, FastAPI, Docker, regulated-domain governance

Manga Lens shipped -- Chrome Web Store

Solo-built Chrome extension integrating **4 AI vision providers** (Claude, GPT-4o, Ollama) via provider-agnostic abstraction. Multi-section capture pipeline, 7-day cache, 29 site selector configs. Full ship discipline: Manifest V3, scoped permissions, Chrome Web Store review.

TypeScript, Manifest V3, multi-provider LLM abstraction

Dream Decoder -- AI Multimodal Platform

Full-stack FastAPI + React/TypeScript app coordinating multimodal AI APIs. Intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass image success **~25-30%** over naive prompt orchestration.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate

EDUCATION

Master's Degree -- Kent State University, Ohio, USA

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages: Python, Java (academic), C++ (IoT), TypeScript, JavaScript, SQL (PostgreSQL, T-SQL, SQLite)

Gen AI & LLM: LLM APIs (Claude, GPT-4o, Gemini, Ollama), LangChain, LlamaIndex, RAG, multi-agent workflows, schema-validated contracts, prompt engineering, structured outputs, evaluation pipelines, guardrails, grounding

ML Foundations: scikit-learn, XGBoost, PyTorch, evaluation harnesses, threshold calibration, walk-forward validation, problem decomposition

Serving / Infra: FastAPI, Flask, Docker, REST APIs, structured logging, observability, Jenkins CI/CD, Grafana

Regulated Domains: HIPAA-conscious governance, de-identification, audit trails, stakeholder-facing limitation docs — transferable to finance / public sector / life sciences

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required